A STRICTLY ENRICHED KANNAN MAP WHICH IS NOT AN ENRICHED CONTRACTION

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Status

This is a candidate full answer, subject to human review, to the open problem in Remark 2(2) of Berinde and Păcurar [1]: find a strictly enriched Kannan mapping which is not an enriched Banach contraction.

Source Question

For a normed space X, the source paper calls $T : X \to X$ an enriched Banach contraction if there are $b \geq 0$ and $0 \leq \theta < b + 1$ such that

$$\|b(x-y) + Tx - Ty\| \leq \theta\|x-y\| \qquad (x, y \in X). \tag{1}$$

The companion paper [2] calls T an enriched Kannan mapping if there are $k \geq 0$ and $0 \leq a < 1/2$ such that

$$\|k(x-y) + Tx - Ty\| \leq a\big(\|x - Tx\| + \|y - Ty\|\big) \qquad (x, y \in X). \tag{2}$$

It is *strictly* enriched Kannan when it is enriched Kannan but not a usual Kannan mapping (the case $k = 0$). The source then asks for such a map which is not an enriched contraction.

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Source paper, p. 4 (Remark 2.2 begins)

□

Remark 2.2. *1) In the particular case $b = 0$, by Theorem 1 we get the classical Banach contraction fixed point theorem (see Banach [2]) in the setting of a Banach space.*

2) Enriched Kannan contractions have been introduced and studied by Berinde and Păcurar in [10]. Although the mapping T in Example 1 (2) is simultaneously an enriched Kannan mapping and an enriched contraction, it is easily seen that the class of enriched Kannan contractions is also independent of the class of enriched Banach contractions,

Source paper, p. 5 (continuation and question)

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due to the fact that, see for example [16], the class of Kannan contractions is independent of the class of Banach contractions.

We shall call a mapping T to be a strictly enriched Kannan mapping if it is not a usual Kannan mapping. So, an open problem would be to find a strictly enriched Kannan mapping which is not an enriched

Proof Intuition

Start with a discontinuous Kannan map S. The affine de-averaging $T = 2S - I$ has averaged map $(I+T)/2 = S$, so the Kannan inequality for S turns verbatim into the enriched Kannan inequality for T with $k = 1$. We choose S piecewise constant, with its jump placed far enough from its fixed point that the Kannan estimate has ample slack. On one whole branch, $T = -I$; evaluating the ordinary Kannan inequality at 1 and -1 forces the forbidden endpoint constant $1/2$. Finally, T retains the jump of S, while every enriched Banach contraction is Lipschitz.

Construction and Full Answer

Define $S, T : \mathbb{R} \to \mathbb{R}$ by

$$S(x) = \begin{cases} 0, & x \leq 2, \\ -\dfrac{1}{3}, & x > 2, \end{cases} \qquad T(x) = 2S(x) - x = \begin{cases} -x, & x \leq 2, \\ -x - \dfrac{2}{3}, & x > 2. \end{cases}$$

Lemma 1. *The map S is a Kannan mapping with constant $1/7$.*

Proof. If x, y lie on the same side of 2, then $Sx = Sy$, so the Kannan inequality is immediate. Suppose, after interchanging x, y if necessary, that $x \le 2 < y$. Then

$$|Sx - Sy| = \frac{1}{3},$$

whereas

$$\frac{1}{7}\big(|x - Sx| + |y - Sy|\big) = \frac{1}{7}\left(|x| + \left|y + \frac{1}{3}\right|\right) > \frac{1}{7} \cdot \frac{7}{3} = \frac{1}{3}.$$

Thus the Kannan inequality holds for every pair, and $1/7 < 1/2$. □

Theorem 1. *The map $T : \mathbb{R} \to \mathbb{R}$ above is a $(1, 1/7)$-enriched Kannan mapping, is not a usual Kannan mapping, and is not an enriched Banach contraction. Hence it solves the open problem in Remark 2(2) of* [1].

Proof. For all $x, y \in \mathbb{R}$, the identity $T = 2S - I$ gives

$$\big|(x - y) + Tx - Ty\big| = 2|Sx - Sy|$$

and

$$|x - Tx| + |y - Ty| = 2\big(|x - Sx| + |y - Sy|\big).$$

Lemma 1, multiplied by 2, is therefore exactly (2) with $k = 1$ and $a = 1/7$. Thus T is enriched Kannan.

It is not a usual Kannan mapping. Indeed, if the ordinary Kannan inequality held with some $c < 1/2$, then at $x = 1$ and $y = -1$ we would have

$$2 = |T1 - T(-1)| \le c\big(|1 - T1| + |-1 - T(-1)|\big) = 4c,$$

which forces $c \ge 1/2$, a contradiction. Consequently T is strictly enriched Kannan.

Finally, every enriched Banach contraction is Lipschitz. In fact, (1) implies

$$\|Tx - Ty\| \le \|b(x - y) + Tx - Ty\| + b\|x - y\| \le (\theta + b)\|x - y\|.$$

But $T(2) = -2$, while

$$\lim_{x \downarrow 2} T(x) = -\frac{8}{3}.$$

Thus T is discontinuous at 2, so it cannot be an enriched Banach contraction. □

Verification Notes

The proof has three independently checkable parts: the only nonzero case in the Kannan estimate for S is the cross-branch case; the pair $(1, -1)$ proves sharp failure of the ordinary Kannan bound; and the jump at 2 contradicts the elementary Lipschitz consequence of the enriched Banach inequality. No numerical or computer-assisted step is used.

The strict inequality $y > 2$ is relevant in Lemma 1: it makes $|y + 1/3| > 7/3$, so the endpoint constant $1/7$ is valid even when $x = 0$. The construction is a self-map of $\mathbb{R}$, which is a Banach space, and all parameters lie in the ranges required by the source definitions.

Novelty Check and Limitations

A bounded search was performed on August 11, 2026. The local run registry, solution, attempt, and proof-gap indexes and the parsed arXiv corpus were searched for arXiv:1909.02382 and the phrases "strictly enriched Kannan", "not an enriched contraction", and close combinations. Web searches used the exact open-problem sentence, the source arXiv id, and combinations of "enriched Kannan", "enriched contraction", "independent", and "counterexample". The searches found later work restating and extending the definitions, but no paper exhibiting or claiming an answer to this exact problem. Novelty is therefore plausible, not certified; a specialist bibliographic review remains appropriate.

The theorem answers only the existence question asked in the source paper. It does not classify strictly enriched Kannan mappings or compare the two classes under additional regularity assumptions.

Human Review Recommendation

Check the cross-branch estimate in Lemma 1, then verify that the authors' convention for "strictly enriched Kannan" is precisely failure of the usual Kannan inequality. If those two points pass, Theorem 1 is a complete explicit answer.

References

[1] V. Berinde and M. Păcurar, *Approximating fixed points of enriched contractions in Banach spaces*, arXiv:1909.02382 (2019).

[2] V. Berinde and M. Păcurar, *Fixed point theorems for Kannan type mappings with applications to split feasibility and variational inequality problems*, arXiv:1909.02379 (2019).